MSBF
Merchant Sales-Based Financing Platform

MERCHANT CREDIT INTELLIGENCE

MODULE 1 • EVIDENCE ARCHITECTURE

Governed Evidence, Contract & Acceptance Lineage

Turn daily business activity into decision-ready evidence.

750
Synthetic merchants
One application per merchant

180 days
Baseline operating history
Daily POS, settlement and deposit evidence

2
Matched scenario alternatives
Baseline • Recession Energy

1,500
Integrated G2 contract rows
Application × scenario • read-only boundary

G0 PHYSICAL FOUNDATION — **G1 GOVERNED RUN CONTROL** — Frozen seed, parameters, run identity and source snapshots govern the entire chain.

Primary evidence progression → • Shared upstream inputs feed later stages; not every SQL join is drawn.

01 MERCHANT & DAILY FOUNDATION

Establish identity, requests and matched histories.

M1.2 Deterministic merchant population

INPUT	Governed seed + reference profiles
LOGIC	Generate merchant entities, industries and relationships.
OUTPUT	**750 merchant profiles; linked entity records**
ROLE	Identity for applications and daily histories.

M1.3 Application & requested structure

INPUT	Merchant + relationship evidence
LOGIC	Construct amount, remittance, total payback and horizon.
OUTPUT	**750 applications; one per merchant**
ROLE	Requested terms are not a financing decision.

M1.4 Daily POS & settlement history

INPUT	Merchant + operating parameters
LOGIC	Generate daily sales, seasonal patterns and settlement behavior.
OUTPUT	**Merchant × date; 180-day POS history**
ROLE	Raw operating evidence, not derived features.

M1.5 Daily deposits & liquidity

INPUT	Merchant + POS settlement histories
LOGIC	Model cash movements, balances, liquidity support and NSF events.
OUTPUT	**Merchant × date; deposit/liquidity history**
ROLE	Missing evidence is not favorable liquidity.

M1.6 Matched operating scenarios

INPUT	Accepted POS + deposit histories
LOGIC	Preserve Baseline; apply the Recession Energy overlay.
OUTPUT	**Merchant × date × scenario; two alternatives**
ROLE	Same merchants and dates; controlled comparison.

02 EVIDENCE, FEATURES & CAPACITY

Separate evidence usability from operating strength.

M1.7 Source quality & confidence

INPUT	Application + seven source families
LOGIC	Assess completeness, freshness, consistency and reconciliation.
OUTPUT	**Application × source family; 5,250 quality records**
ROLE	Evidence confidence remains separate from credit risk.

M1.8 Verification, fraud & continuity

INPUT	Source-quality, identity and operating evidence
LOGIC	Evaluate verification, fraud and processor-continuity checks.
OUTPUT	**4,500 atomic checks + 750 application summaries**
ROLE	Separate evidence dimensions; preserve review and blocked paths.

M1.9 As-of cash-flow features

INPUT	Matched histories + quality/verification evidence
LOGIC	Compute trailing operating features at the decision-time cutoff.
OUTPUT	**1,500 wide snapshots + 54,000 long-form feature values**
ROLE	Source hashes and usability accompany application/scenario features.

M1.10 Obligations & residual capacity

INPUT	Requested terms + features + obligation evidence
LOGIC	Assess payment coverage, residual cash flow and post-financing liquidity.
OUTPUT	**Application × scenario; capacity and affordability evidence**
ROLE	Sales are not free cash flow. Preserve evidence and review gates.

03 RESILIENCE, RISK & ECONOMICS

Connect affordability, comparative risk and economic viability.

M1.11 Operating resilience

INPUT	Cash-flow features + capacity snapshots
LOGIC	Build operating archetypes and components; enforce matched stress floors.
OUTPUT	**1,500 resilience snapshots + 7,500 component records**
ROLE	Explain revenue, liquidity, burden, continuity and confidence.

M1.12 Integrated comparative risk

INPUT	Verification + capacity + resilience evidence
LOGIC	Persist seven risk components, weighted score, reasons and floors.
OUTPUT	**1,500 risk snapshots + 10,500 component records**
ROLE	Synthetic comparative proxy; not a calibrated probability of default.

M1.13 Exposure, recovery & loss

INPUT	Request + capacity + risk + recovery evidence
LOGIC	Build exposure paths, timing weights and supported recovery assumptions.
OUTPUT	**Application/scenario snapshots + daily exposure paths**
ROLE	Explicit exposure denominator; comparative loss, not CECL.

M1.14 Unit economics & contribution

INPUT	Request/channel + capacity + comparative loss
LOGIC	Separate revenue, costs, loss, capital charge, contribution and hurdle.
OUTPUT	**1,500 economic snapshots + 21,000 component records**
ROLE	Conditional-if-booked economics; modeled contribution, not audited profit.

04 CONTRACTS & INDEPENDENT G2 ASSURANCE

Two distinct contracts → read-only integration → end-to-end acceptance

M1.15 Application consumption contract

`M1_APPLICATION_CONSUMPTION v1`

1,500 latest + 1,500 immutable archive rows

750 paired comparisons • scenario-aware evidence

Accepted M1.7–M1.14 outputs; preserve hashes and lineage.

M1.16 Acquisition attribution & cost companion

`M1_ACQUISITION_CONSUMPTION v1`

750 latest + 750 immutable archive rows

Application-level • scenario-invariant companion

Attribution, allocated cost and M1.14 overlap bridge; additive only.

M1.17 End-to-end QA & G2 acceptance

`G2_M1_CONTRACT = PASS`

Sources + contracts + archives + evidence

Read-only integration • independent reconciliation

Assure boundary, identity and nonregression; retain provenance.

`M1_G2_CONSUMPTION_BUNDLE v1` — **1,500-row integrated boundary → Module 2 decisioning → G3-v2 → Power BI analytical consumption**

Evidence. Decisions. Accountability.

Independent portfolio • Andrew R. Goad

Deterministic synthetic evidence • Non-production • No real funds